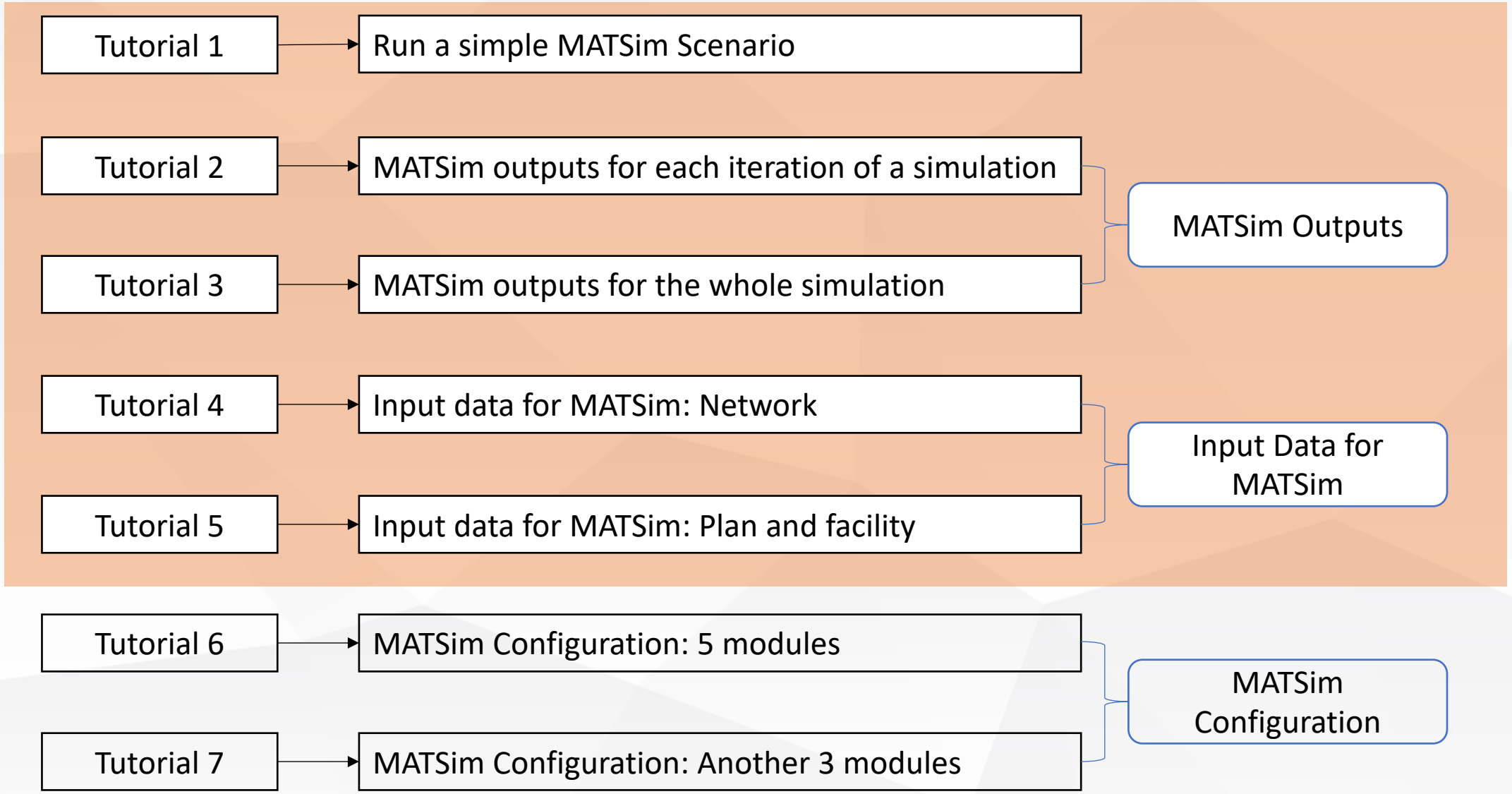
LSGI3213: Intelligent Transportation Systems

Group Discussion on Group Project (B)

Subject Leader: Chengxiang (Tony) ZHUGE (chengxiang.zhuge@polyu.edu.hk)

Tutor: Ruichen MA (ruichen.ma@connect.polyu.hk)

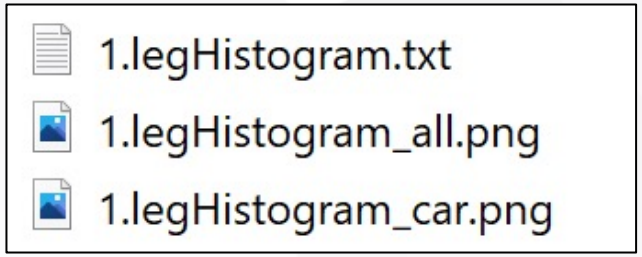
Review of First 5 Tutorials



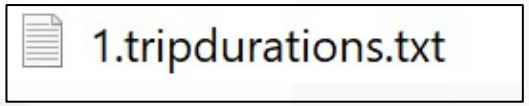
Output files stored in the iteration

The following output files are generated for each iteration:

- **Leg Histogram**



- **Trip Durations**



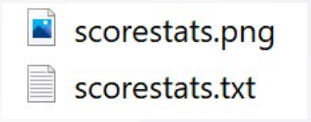
The following output files are generated for Specific iterations (every 10 iterations by default) :

- **Link States**

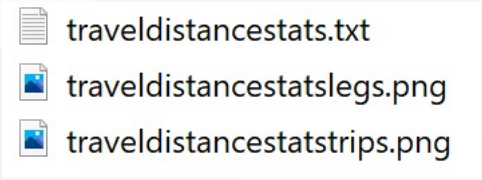


Output files related to the whole simulation

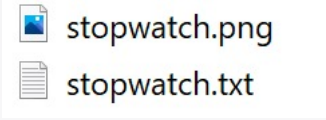
- **Score Statistics**



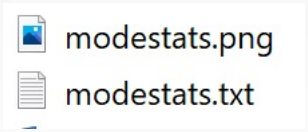
- **Travel Distance Statistics**



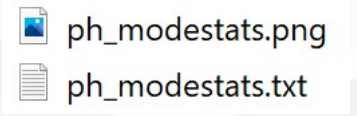
- **Stopwatch**



- **Mode statistics**



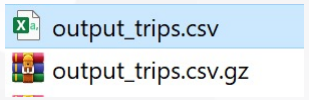
- **Passenger hours travelled per Mode**



- **Passenger kilometers travelled per Mode**



- **Trip information**

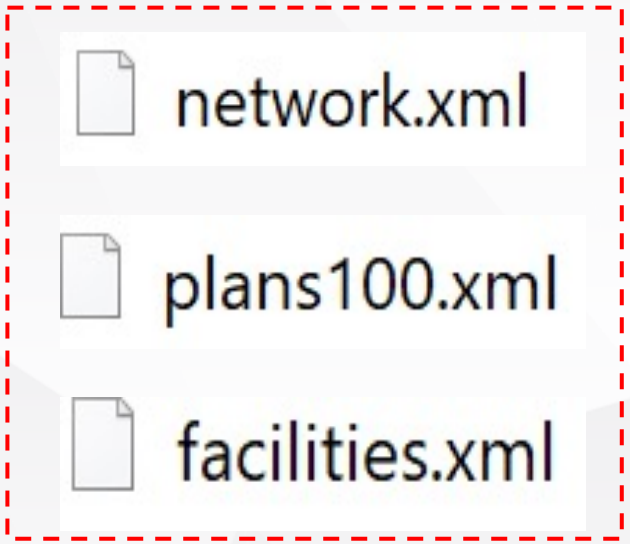


Overview of the input data

Three Typical Types

- **Network Data**
- **Population Data** (or **Plan Data**)
- **Facility Data**

In the **“equil”** scenario



Network.xml

1. **Network Data**
2. Population Data (or Plan Data)
3. Facility Data

Introduction to network.xml

Nodes

```
<nodes>
    <node id="1" x="-20000" y="0"/>
    <node id="2" x="-15000" y="0"/>
    <node id="3" x="-865" y="5925"/>
    <node id="4" x="-2498" y="4331"/>
    <node id="5" x="-3829" y="3215"/>
    <node id="6" x="-4698" y="1711"/>
    <node id="7" x="-5000" y="0"/>
    <node id="8" x="-4698" y="-1711"/>
    <node id="9" x="-3829" y="-3215"/>
    <node id="10" x="-2498" y="-4331"/>
    <node id="11" x="-865" y="-5925"/>
    <node id="12" x="0" y="0"/>
    <node id="13" x="5000" y="0"/>
    <node id="14" x="5000" y="-10000"/>
    <node id="15" x="-20000" y="-10000"/>
</nodes>
```

Links

```
<links capperiod="01:00:00">
    <link id="1" from="1" to="2" length="10000.00" capacity="36000" freespeed="27.78" permlanes="1" />
    <link id="2" from="2" to="3" length="10000.00" capacity="3600" freespeed="27.78" permlanes="1" />
    <link id="3" from="2" to="4" length="10000.00" capacity="3600" freespeed="27.78" permlanes="1" />
    <link id="4" from="2" to="5" length="10000.00" capacity="3600" freespeed="27.78" permlanes="1" />
    <link id="5" from="2" to="6" length="10000.00" capacity="3600" freespeed="27.78" permlanes="1" />
    <link id="6" from="2" to="7" length="10000.00" capacity="3600" freespeed="27.78" permlanes="1" />
    <link id="7" from="2" to="8" length="10000.00" capacity="3600" freespeed="27.78" permlanes="1" />
    <link id="8" from="2" to="9" length="10000.00" capacity="3600" freespeed="27.78" permlanes="1" />
    <link id="9" from="2" to="10" length="10000.00" capacity="3600" freespeed="27.78" permlanes="1" />
    <link id="10" from="2" to="11" length="10000.00" capacity="3600" freespeed="27.78" permlanes="1" />
    <link id="11" from="3" to="12" length="5000.00" capacity="1000" freespeed="27.78" permlanes="1" />
    <link id="12" from="4" to="12" length="5000.00" capacity="1000" freespeed="27.78" permlanes="1" />
    <link id="13" from="5" to="12" length="5000.00" capacity="1000" freespeed="27.78" permlanes="1" />
    <link id="14" from="6" to="12" length="5000.00" capacity="1000" freespeed="27.78" permlanes="1" />
    <link id="15" from="7" to="12" length="5000.00" capacity="1000" freespeed="27.78" permlanes="1" />
    <link id="16" from="8" to="12" length="5000.00" capacity="1000" freespeed="27.78" permlanes="1" />
    <link id="17" from="9" to="12" length="5000.00" capacity="1000" freespeed="27.78" permlanes="1" />
    <link id="18" from="10" to="12" length="5000.00" capacity="1000" freespeed="27.78" permlanes="1" />
    <link id="19" from="11" to="12" length="5000.00" capacity="1000" freespeed="27.78" permlanes="1" />
    <link id="20" from="12" to="13" length="10000.00" capacity="36000" freespeed="27.78" permlanes="1" />
    <link id="21" from="13" to="14" length="10000.00" capacity="36000" freespeed="27.78" permlanes="1" />
    <link id="22" from="14" to="15" length="35000.00" capacity="36000" freespeed="27.78" permlanes="1" />
    <link id="23" from="15" to="1" length="10000.00" capacity="36000" freespeed="27.78" permlanes="1" />
</links>
```

In the **“equil”** scenario, the network is composed of **15** nodes and **23** links.

Via

For visualization

Plans100.xml

1. Network Data
2. **Population Data (or Plan Data)**
3. Facility Data

Introduction to plan100.xml

In the **“equil”** scenario, the plans100.xml file contains **100** agents and their daily plans.

The plan data is organized in a hierarchical structure

```
<person id="1">
    <plan>
        <act type="h" x="-15050" y="-150" link="1" end_time="06:00" />
        <leg mode="car">
            <route>2 7 12</route>
        </leg>
        <act type="w" x="4950" y="-150" link="20" dur="00:10" />
        <leg mode="car">
            <route> </route>
        </leg>
        <act type="w" x="4950" y="-150" link="20" dur="03:30" />
        <leg mode="car">
            <route>13 14 15 1</route>
        </leg>
        <act type="h" x="-15050" y="-150" link="1" />
    </plan>
</person>
```

Person 1

```
<person id="2">
    <plan>
        <act type="h" x="-15050" y="-150" link="1" end_time="05:59" />
        <leg mode="car">
            <route>2 7 12</route>
        </leg>
        <act type="w" x="4950" y="-150" link="20" dur="02:30" />
        <leg mode="car">
            <route>13 14 15 1</route>
        </leg>
        <act type="h" x="-15050" y="-150" link="1" />
    </plan>
</person>
```

Person 2

```
<person id="3">
    <plan>
        <act type="h" x="-15050" y="-150" link="1" end_time="05:58:30" />
        <leg mode="car">
            <route>2 7 12</route>
        </leg>
        <act type="w" x="4950" y="-150" link="20" dur="01:30" />
        <leg mode="car">
            <route>13 14 15 1</route>
        </leg>
        <act type="h" x="-15050" y="-150" link="1" />
    </plan>
</person>
```

Person 3

Facility.xml

1. Network Data
2. Population Data (or Plan Data)
3. **Facility Data**

Introduction to facilities.xml

identifier

```
<!-- ====================================================================== -->

    <facility id="1" x="-17500.0" y="0.0">
        <activity type="h">
        </activity>

        <activity type="w">
        </activity>

    </facility>

<!-- ====================================================================== -->

    <facility id="10" x="-7932.5" y="-2962.5">
        <activity type="h">
        </activity>

        <activity type="w">
        </activity>

    </facility>

<!-- ====================================================================== -->

    <facility id="11" x="-432.5" y="2962.5">
        <activity type="h">
        </activity>

        <activity type="w">
        </activity>

    </facility>

<!-- ====================================================================== -->
```

Facility 1

Facility 10

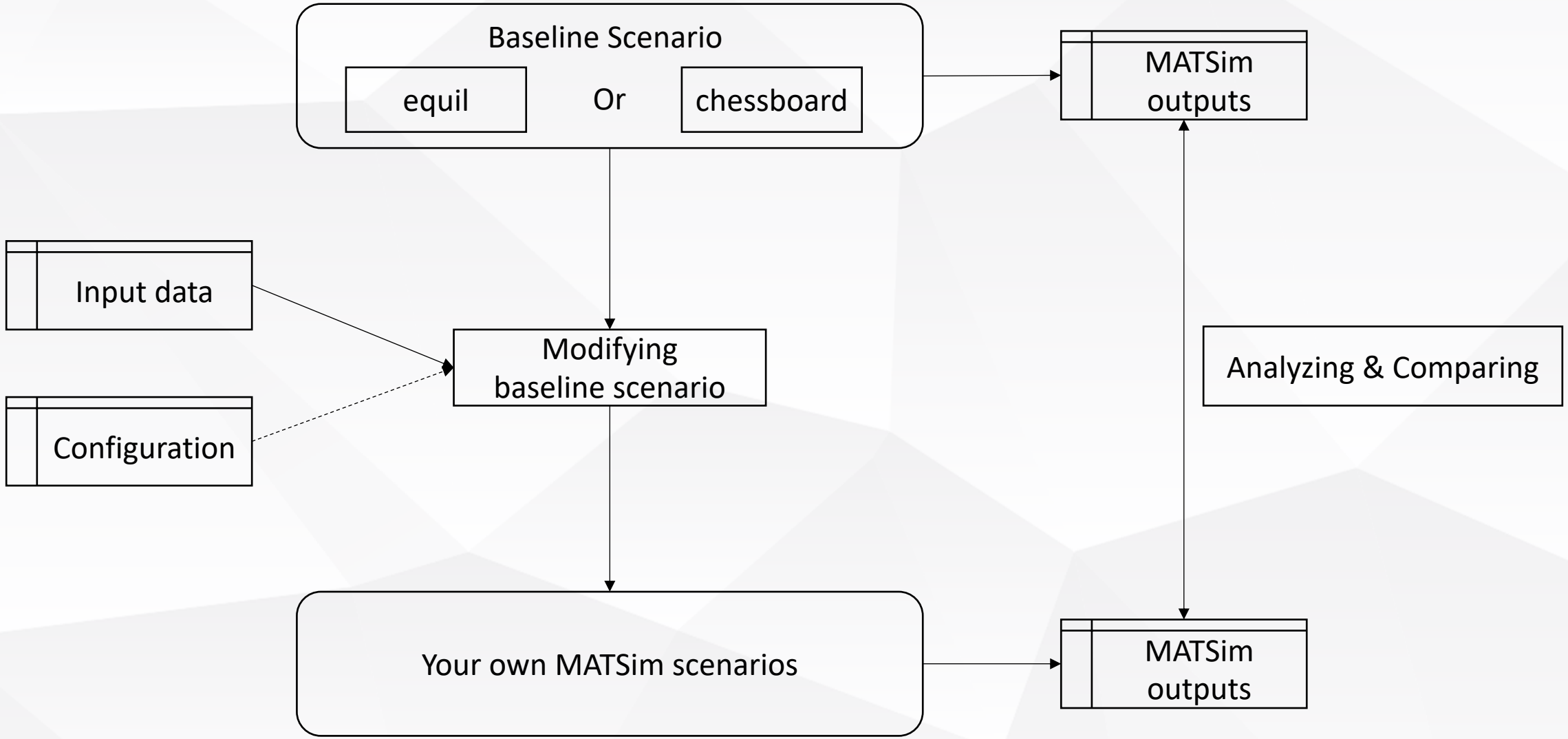
Facility 11

home

work

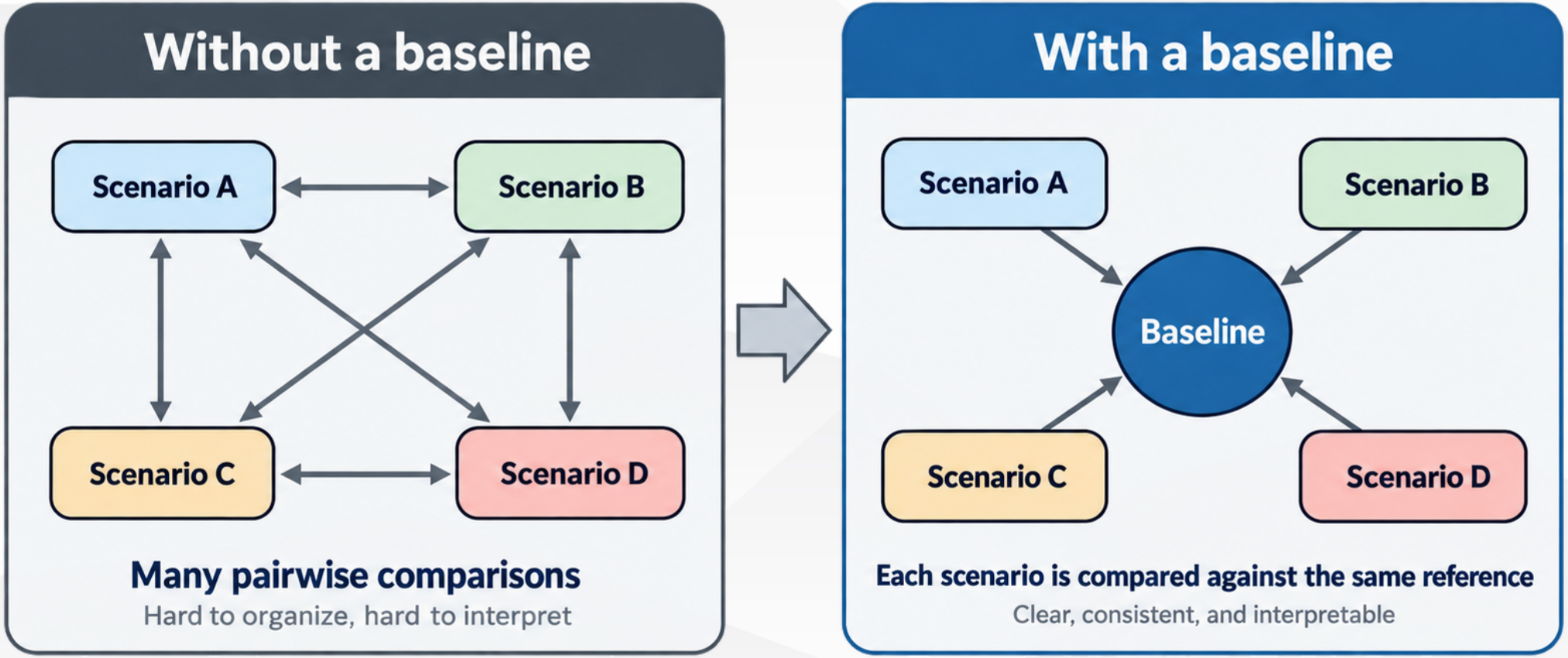
In the **“equil”** scenario, there are **23** facilities.

Group discussion on Group Project



Group discussion on Group Project

- A baseline is a standard reference scenario that remains unchanged and gives us a consistent basis for comparison.



Discuss with your group members

Questions

If you have any technical problem with MATSim or our first 5 tutorials, you may discuss with your group members firstly or turn to the Tutors for help.

- Ruichen MA (Email: ruichen.ma@connect.polyu.hk)

You may wish to copy Dr. Zhuge in when you send the Tutors an email.

- Chengxiang ZHUGE: chengxiang.zhuge@polyu.edu.hk